

GRAIN DIRECTION

SAME SIZE ARTWORK
LEAFLET: 225 mm x 270 mm
FOLDING SIZE: 38 mm x 135 mm



135 mm

SPACE FOR PHARMACODE

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For the use only of a Registered Medical Practitioner
or a Hospital or a Laboratory

Budesma Pressurised Inhalation BP 200 mcg

CFC FREE

200 mcg /Actuation

300 Metered Doses

Composition

Each actuation delivers:

Budesonide BP 200 mcg

Description

White colour suspension filled in Aluminium aerosol can with silver ferrule valve provided with actuator suitable for oral inhalation, when actuated a metered spray is produced.

Pharmacology

Budesonide is a potent glucocorticoid that when given systemically has standard glucocorticoid activity and binds with high affinity to the glucocorticoid receptor. The drug was developed primarily for topical use. Budesonide, if given systemically, has catabolic and antianabolic effect on proteins in peripheral tissues; it causes resistance to insulin and impairs the peripheral utilization of glucose as well as causing increased calcium loss from bone via the kidney. However budesonide depends for its effects on the local anti-inflammatory and immunosuppressive activity. It has been shown in animal studies to have a high ratio of topical to systemic activity compared with other corticosteroids including beclomethasone dipropionate, flunisolide, flucinolone acetoneide, and triamcinolone acetoneide. Results of a study of hamsters with induced ileitis suggested that the anti-inflammatory effect of budesonide was due to local rather than systemic effects. Inhaled glucocorticoids reduce the number of inflammatory cells such as eosinophils, lymphocytes and mast cells and restore airway epithelial integrity in bronchial biopsy specimens obtained from mild asthmatic patients. Inhaled budesonide also reduces indices of eosinophil activation in asthma. These effects are likely to result from inhibition of transcription of several cytokines that are over expressed in asthma, in particular the interleukins IL-3, IL-4 and IL-5 and granulocyte macrophage colony stimulating factor (GM-CSF), especially from activated T cells. Glucocorticoids also inhibit plasma exudation through the endothelial barrier of the bronchial vasculature and therefore, their administration leads to a reduction in airway edema. Budesonide inhibits phospholipase A2 activity and by this means reduces the formation of prostaglandins and leukotrienes in the airway.

Pharmacokinetics

The preferred analytical method is by high performance liquid chromatography in series with radioimmunoassay, which give a sensitivity of 1 microgram per litre. There is no interference from endogenous steroids.

Absorption from the gastrointestinal tract is probably complete. Oral bio-availability is approximately 10 % indicating high pre-systemic metabolism by liver. Studies with 3H-budesonide show no metabolism in lung or nasal mucosa. Peak plasma levels are reached in about 30 min after nasal application; after inhalation within 5-10 mins; after rectal administration within 1.5 hour. Rectal administration of 2 mg budesonide gave a mean maximal plasma concentration of 3 nmol/l (range 1-9 nmol/l). Plasma half life is approximately 2 hours and the plasma clearance ratio is 0.9-1.41/min. The metabolites have minimal biological activities. Inhalation gives a high plasma concentration, 2 nmol/l after 500 microgram budesonide, resembling that obtained after intravenous administration. This suggests effective lung deposition and minimal biotransformation in the lung, since poor lung deposition would give a delayed plasma peak as with oral administration. The volume of distribution after intravenous administration is 301+/- 421 and there is little or no entry into the CNS. Plasma protein binding is 86-90% mainly to albumin and not to transcortin.

Budesonide is extensively distributed. Tissue distribution has been studies in mouse which showed some placental transfer, but it is not known for humans. No data in concentrations in breast milk in humans is available. Metabolism is more rapid in small children. Metabolic transformation of budesonide in liver is not affected by drugs which inhibit some forms of cytochrome P450 (e.g. cimetidine).

Oral absorption	~100%
Pre-systemic metabolism	90%
Plasma half-life mean	2 hours
Volume of distribution	43 l/kg
Plasma protein binding	86-90%

Pharmacodynamics

Budesonide is an anti-inflammatory corticosteroid that exhibits potent glucocorticoid activity and weak mineralocorticoid activity. The precise mechanism of corticosteroid action on inflammation in asthma is not known. Corticosteroids have been shown to have a wide range of inhibitory activities against multiple cell types (mast cells, eosinophils, macrophages and lymphocytes) and mediators (e.g. histamine, eicosanoids, leukotrienes and cytokines) involved in allergic and non-allergic mediated inflammation.

Indications

Bronchial asthma

Dosage and Administration

For oral inhalation.

The dosage of budesonide is individual. Initially at the beginning of therapy, for therapy during severe asthma or when scaling down or withdrawing oral corticosteroids, the dosage should be:

Children aged 2-7 years:

200-400 micrograms per day in two divided doses

Children aged 7 years and older:

200-800 micrograms per day in two divided doses

Adults:

200-1600 micrograms per day in two divided doses

For maintenance treatment administration twice daily morning and evening is sufficient. The maintenance dose should be the lowest possible. Following a single dose, an effect may be expected after a few hours. The full therapeutic effect is only achieved after a few weeks of treatment.

Contraindications

Budesonide Pressurised Inhalation is contra-indicated in patients with a history of hypersensitivity to budesonide or any of the components of the drug product.

Warning and Precautions

Budesonide Pressurised Inhalation is not designed to relieve acute symptoms for which an inhaled short-acting bronchodilator is required. Patients should be made aware of the prophylactic nature of the therapy with inhaled budesonide and it should be taken regularly. Systemic effects of inhaled steroids may occur particularly at high doses for prolonged periods. These effects are much less likely to occur than with oral steroids. Possible systemic effects include adrenal suppression, growth retardation in children and adolescents, decrease in bone density, and cataract and glaucoma in the eyes. It is therefore important that the dose of inhaled corticosteroid is titrated to the lowest at which effective control of asthma is achieved. It is recommended that the height of children receiving prolonged therapy with inhaled steroids be regularly monitored. As with all inhaled corticosteroids, special care is necessary in patients with active or quiescent pulmonary tuberculosis. Children on immunosuppressant drugs are more susceptible to infections than healthy children. Chicken pox and measles can have a more serious and even fatal course in children on immunosuppressant corticosteroids. Replacement of systemic steroid therapy with inhaled therapy sometimes unmasks allergies such as allergic rhinitis or eczema previously controlled by the systemic drug. These should be symptomatically treated with anti histamine and/or topical steroid preparations.

For transfer of patients being treated with oral steroids:

The transfer of oral steroid-dependent patients to Budesonide Pressurised Inhalation and their subsequent management needs special care as recovery from impaired adreno-cortical function due to prolonged systemic steroid therapy may take a considerable time. Gradual withdrawal of the systemic steroid is recommended and in these patients, the possibility of residual impaired adrenal response should always be considered in emergency and elective situations likely to produce stress and appropriate corticosteroid treatment considered.

Drug Interactions

The metabolism of budesonide is primarily mediated by CYP3A4, one of the cytochrome p450 enzymes. Inhibitors of this enzyme e.g ketoconazole and itraconazole therefore can increase systemic exposure to budesonide. Other potent inhibitors are also likely to markedly increase plasma levels of budesonide.

(i) Products containing corticosteroids other than Beclomethasone:

a) Interactions with Other Medicaments:

Co-treatment with CYP3A inhibitors, including cobicistat-containing products, is expected to increase the risk of systemic side-effects. The combination should be avoided unless the benefit outweighs the increased risk of systemic corticosteroid side-effects, in which case patients should be monitored for systemic corticosteroid side-effects.

Pregnancy

Budesonide has been labelled as pregnancy category B. The possibility of fetal harm is remote if the drug is used during pregnancy. Nevertheless, because the studies in humans cannot rule out the possibility of harm, budesonide should be used during pregnancy only if clearly needed.

Lactation

It is not known whether budesonide is distributed in milk. However, other steroids are distributed in milk hence because of the potential of serious adverse reactions from corticosteroids in nursing infants, a decision should be made whether to discontinue nursing or budesonide taking into account the importance of the drug to the woman.

Undesirable effects

Clinical trials, literature reports and post-marketing experience suggest that the following adverse drug reactions may occur:

Common (>1/100, <1/10):

- Mild throat irritation
- Candida infection in the oropharynx
- Hoarseness of voice
- Coughing

Rare (>1/10000, <1/1000):

- Nervousness, restlessness
- Behavioural disturbances, depression
- Skin bruising
- Immediate and delayed hypersensititv reactions including rash/contact dermatitis/urticaria/ angioedema and bronchospasm

Candida infection in the oropharynx is due to drug deposition. Advising the patient to rinse the mouth with water after each dosing will minimise the risk. As with other inhalation therapy, paradoxical bronchospasm may occur in very rare cases.

Systemic effects of inhaled corticosteroids may occur, particularly at high doses prescribed for prolonged periods. These effects are much less likely to occur than with oral corticosteroids. Possible systemic effects include adrenal suppression, growth retardation in children and adolescents, decrease in bone density, and cataract and glaucoma in the eyes. The effect is probably dependent on dose, exposure time, concomitant and previous steroid exposure and individual sensitivity.

Overdose

Acute overdose with budesonide inhaler may lead to temporary suppression of adrenal function. This does not necessitate emergency action being taken. In these patients treatment with budesonide by inhalation should be continued at a dose sufficient to control the asthma. Adrenal function returns in a few days and can be verified by measuring plasma cortisol. Chronic overdosage may lead to adrenal suppression. Monitoring of adrenal reserve may be indicated. Treatment with inhaled budesonide should be continued at a dose sufficient to control asthma.

Storage

Store below 30 °C. Do not freeze.
Keep out of reach of children.

Route of administration: Oral Inhalation

Packing/pack size

CFC-free metered dose inhaler preparation of Budesonide that delivers 200 mcg of the drug with every actuation.

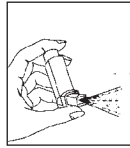
Presentation

300 Metered Doses

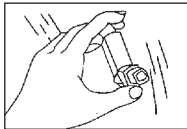
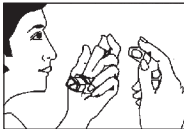
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INSTRUCTIONS FOR PATIENT FOR USING THE INHALER WITHOUT SPACER

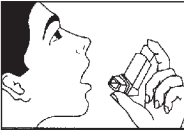
Important: Follow instructions carefully.



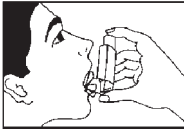
1



2



3



4



5

PE00000 MY

ICONGRAPHICS CODE:

PANTONE SHADE



PANTONE
BLACK
PROCESS C

**Supersedes
Artwork Code:**

PHARMACODE:



For Children:
Children should use the inhaler under adult supervision, as instructed by the Physician. (see pic no. 8)



8

Clean the inhaler at least once a day.

- Discard the canister after using the labeled number of inhalation.**



Important: Follow instructions carefully.

Testing your inhaler:



A spacer is an extra plastic tube that fits on the end of the mouthpiece. It holds the asthma medicine that is released from the canister briefly, so that you have a better chance of inhaling it properly.



3

4

5

6

It's important to care for your metered dose inhaler and your spacer carefully and correctly. This keeps everything working and helps you to get the right amount of medicine with each puff. Be sure to:

7

Continue to breathe in as slowly and as deeply as you can manage. Then, take the spacer out of your mouth.

PE00000 MY

PHARMACODE: